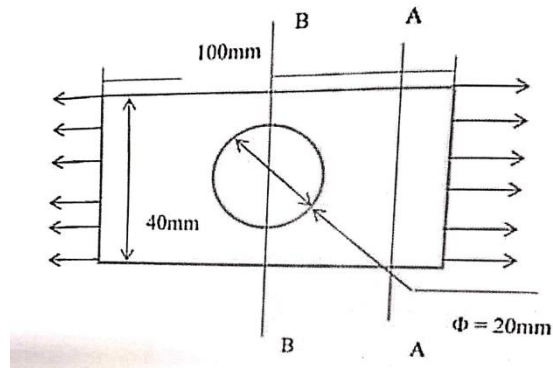


Experiment No. 3

1. **Title:** Plate/Shell Element – Structural Linear
2. **Aim:** To perform finite element analysis of a plate structure using ANSYS APDL plate/(shell) elements in order to determine deformation, stress distribution, and reaction forces under applied loading conditions.

3. Problem Statement:

A plate of length 100 mm, width 40 mm and with thickness 10 mm subjected to tension of 900 N and this force is applied as uniformly distributed force ($E = 210 \text{ GPa}$ and $\nu = 0.3$). Determine normal stresses, shear stresses, Equivalent (Von-Mises) Stress, principal stresses and deformation using ANSYS APDL



4. Solve given problem using basic steps of FEA (Manual method)

$$\sigma_{\max} = \sigma_{\text{avg}} \times K_c$$

$$\sigma_{\text{avg}} = F / A$$

$$= F / (b - d)t$$

$$= 900 / (40 - 20) \times 10$$

$$\sigma_{\text{avg}} = 4.5 \text{ N/mm}^2$$

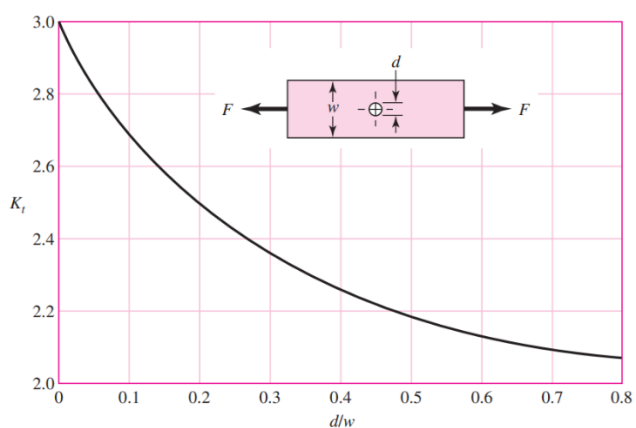
$$d / b = 20 / 40 = 0.5$$

$$\therefore K_c \approx 2.15$$

$$\sigma_{\max} = \sigma_{\text{avg}} \times K_c$$

$$= 4.5 \times 2.15$$

$$\sigma_{\max} = 9.67 \text{ N/mm}^2$$



5. Steps In Ansys to solve problems

For finding maximum stress in Ansys we use pressure instead of force. Pressure on the plate can be calculated as follows:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Pressure} = 900/(40 * 10) = 2.25 \text{ N/mm}^2$$

a) Preprocessing

1. ANSYS Main Menu → Preferences → Select → STRUCTURAL → OK.
2. Element Type – Add/Edit/Delete → Add → Solid → Quad 4 node 182 → OK.
3. Options – Element Behaviour (K3) Plate strs w/thk → OK.
4. Real Constants – Add/Edit/Delete → Set 1 → Add → Type 1 PLANE182 → OK →
5. Real Constant Set No. = 1 → THK = 10.
6. Material Props – Material Models → Structural → Linear → Elastic → Isotropic → Young's Modulus EX = 2e5, PRXY = 0.3 → Close.
7. Modeling – Create → Areas → Rectangle → By 2 Corners → WP X = 0, WP Y = 0, Width = 50, Height = 20 → OK (Considering symmetry of the geometry).
8. Modeling – Create → Areas → Circle → Solid Circle → WP X = 0, WP Y = 0, Radius = 10 → OK.
9. Modeling – Operate → Booleans → Subtract → Areas → Select Rectangle → Apply → Select Circle → Apply → OK.
10. Meshing – Mesh Tool → Select Smart Size using slider = Fine (2) → Click Mesh button in Mesh Tool → Select geometry → Apply → Close Mesh Tool.
11. Loads – Define Loads → Apply → Structural → Displacement → Symmetry B.C. → On Lines → Select symmetry lines → OK.
12. Loads – Define Loads → Apply → Structural → Pressure → On lines → Select right edge of geometry → Apply → VALUE Load PRES = -2.25.

b) Solution

13. Solution → Solve → Current Load Step → OK.

c) Post-Processing

• Total Deformation (Deflection Plot)

14. General Postproc → Plot Results → Deformed Shape → Def + Undeformed → OK.

OR

15. General Postproc → Plot Results → Contour Plot → Nodal Solution → DOF Solution → Sum of Displacement (USUM) → OK.

• Normal Stress Distribution

16. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → SX (X-direction stress) → OK

17. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → SY (Y-direction stress) → OK

• Shear Stress

18. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → SXY (Shear Stress) → OK.

• Equivalent (Von-Mises) Stress: Most important for failure analysis.

19. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → SEQV (von-Mises Stress) → OK.

• Principal Stresses: Maximum and minimum stresses.

20. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → S1 (Max Principal)

21. General Postproc → Plot Results → Contour Plot → Nodal Solution → Stress → S1 (Max Principal)

• Reaction Forces at Supports: To verify boundary conditions.

22. General Postproc → List Results → Reaction Solution → All Items → OK.

d) Result

	Finite Element Method (Manual Calculation)	Finite Element Method (Ansys APDL)
Maximum Stress		

e) Conclusion

The finite element analysis of the plate using ANSYS APDL plate/shell elements was successfully performed under a uniformly distributed tensile load of 900 N. The manually calculated maximum normal stress (**9.67 N/mm²**) closely matched the ANSYS results, confirming the accuracy of the model. Stress distribution, deformation, von Mises stress, and principal stresses were effectively captured, demonstrating the reliability of linear structural FEA for plate analysis.